The Mathematical Tools of Quantum Mechanics (Cohen–Tannoudji, Ch. II) Section B. State space and Dirac notation

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Outline

- A. One-particle wave function space
- B. State space and Dirac notation
- C. Representations in the state space
- D. Eigenvalue equations and observables
- E. Two key examples: $|\boldsymbol{r}\rangle$ and $|\boldsymbol{p}\rangle$; operators $\boldsymbol{R}$, $\boldsymbol{P}$
- F. Tensor products and applications

From wave functions to state vectors

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- Associate to each physical state a ket $|\psi\rangle$:
  $\psi(\boldsymbol{r}) \in \mathcal{F} \iff |\psi\rangle \in \mathcal{E}_{\boldsymbol{r}}$
- $\psi(\boldsymbol{r})$ will be interpreted as the set of the components of the ket $|\psi\rangle$ in a particular basis.

Bras, kets, and the dual space

Dual space $\mathcal{E}^*$: Lets consider first a linear functional $\mathcal{X}$:

$$\begin{aligned} \mathcal{E} &\xrightarrow{\chi} \mathbb{C} \\ |\psi\rangle &\longmapsto \chi(|\psi\rangle) \end{aligned}$$

The set of linear functionals defined on the kets $|\psi\rangle \in \mathcal{E}$ constitutes a vector space called dual space of $\mathcal{E}$, symbolized by $\mathcal{E}^*$.

- Any vector of $\mathcal{E}^*$ is called a bra vector, or bra: $\chi(|\psi\rangle) = \langle\chi|\psi\rangle$

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- Any vector of $\mathcal{E}^*$ is called a bra vector, or bra: $\chi(|\psi\rangle) = \langle\chi|\psi\rangle$
- To every ket corresponds a bra: the ket $|\phi\rangle$ enables to define a linear functional associating for each ket $|\psi\rangle$ a complex number equal to the scalar product $(|\phi\rangle, |\psi\rangle)$. $\langle\phi|$ is this linear functional: $\langle\phi|\psi\rangle = (|\phi\rangle, |\psi\rangle)$

Bras, kets, and the dual space

- Braket correspondence is antilinear with respect to the first vector:

$$\langle \lambda_1 \phi_1 + \lambda_2 \phi_2 \mid \psi \rangle = \lambda_1^* \langle \phi_1 \mid \psi \rangle + \lambda_2^* \langle \phi_2 \mid \psi \rangle.$$

The bra associated with the ket $|\lambda_1 \phi_1 + \lambda_2 \phi_2\rangle$ is the bra

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- If $\lambda \in \mathbb{C}$ and $|\psi\rangle$ is a ket, then $\lambda|\psi\rangle$ is also a ket (written as $|\lambda\psi\rangle$). Since the correspondence between a bra and a ket is antilinear,

$$\boxed{\langle \lambda\psi| = \lambda^* \langle \psi|}.$$

Dirac notation for the scalar product

We have two distinct notations for the scalar product of $|\psi\rangle$ by $|\phi\rangle$: $(|\phi\rangle, |\psi\rangle)$ or $\langle\phi|\psi\rangle$. From now on, we will use the Dirac notation for the scalar product: $\langle\phi|\psi\rangle$

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- Linear in second: $\langle\phi \mid \lambda_1\psi_1 + \lambda_2\psi_2\rangle = \lambda_1\langle\phi \mid \psi_1\rangle + \lambda_2\langle\phi \mid \psi_2\rangle$

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- Positive: $\langle\psi \mid \psi\rangle \in \mathbb{R}_{\geq 0}, \quad \langle\psi \mid \psi\rangle = 0 \iff |\psi\rangle = 0.$

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- The commutator $[A, B] = AB - BA$
- Let $|\phi\rangle$ and $|\psi\rangle$ be two kets. Matrix element of A between $|\phi\rangle$ and $|\psi\rangle$: $\langle\phi|(A|\psi\rangle)$. It is a number (linear on $|\psi\rangle$ and antilinear on $|\phi\rangle$)

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- Writing $|\psi\rangle\langle\phi|$ represents an operator. If you apply it to an arbitrary ket, it gives another ket.
- The order of symbols is of critical importance. Only complex numbers (e.g. λ) can be moved freely:

$$|\psi\rangle\lambda = \lambda|\psi\rangle$$
$$\langle\psi|\lambda = \lambda\langle\psi|$$
$$A\lambda|\psi\rangle = \lambda A|\psi\rangle \ (A \text{ is operator})$$
$$\langle\phi|\lambda|\psi\rangle = \lambda\langle\phi|\psi\rangle = \langle\phi|\psi\rangle\lambda$$

Projector onto a ket $|\psi\rangle$

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P_ψ acting on an arbitrary ket $|\phi\rangle$ gives a ket proportional to $|\psi\rangle$.

Geometrical significance of P_ψ: it is the orthogonal projection onto the ket $|\psi\rangle$.

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$$P_\psi^2 = P_\psi P_\psi = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = P_\psi$$

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In this expression, $\langle\psi|\psi\rangle$ is a number, which is equal to 1 . Therefore,

$$P_\psi^2 = P_\psi$$

Projector onto a subspace

Let $|\phi_1\rangle, \ldots, |\phi_q\rangle$ be q normalized vectors orthogonal to each other:

$$\langle \phi_i | \phi_j \rangle = \delta_{ij} \quad (i, j = 1, \ldots, q)$$

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$$P_q = \sum_{i=1}^{q} |\phi_i\rangle\langle\phi_i|$$

Calculate P_q^2:

$$P_q^2 = \sum_{i=1}^{q}\sum_{j=1}^{q} |\phi_i\rangle\langle\phi_i|\phi_j\rangle\langle\phi_j| = \sum_{i=1}^{q} |\phi_i\rangle\langle\phi_i| = P_q$$

Thus, P_ϕ is a projector.

Action of a Linear Operator on a Bra

Reminder

We already defined the action of a linear operator A on kets:

$$A\,|\psi\rangle\,.$$

Action of a Linear Operator on a Bra

New idea

For fixed $\langle\phi|$ and A, the number $\langle\phi| A |\psi\rangle$ depends linearly on $|\psi\rangle$.
Hence it defines a new linear functional on kets.

Action of a Linear Operator on a Bra

$$\boxed{(\langle\phi|A)|\psi\rangle \;=\; \langle\phi|\big(A|\psi\rangle\big)}$$

This relation *defines* the bra that results from the action of A on $\langle\phi|$. We denote it by $\langle\phi|A$ (a new bra).

Linearity of the Action on Bras

Let $\langle\chi| = \lambda_1 \langle\phi_1| + \lambda_2 \langle\phi_2|$.

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$$\boxed{\langle\chi| A = \lambda_1 \langle\phi_1| A + \lambda_2 \langle\phi_2| A}$$

The correspondence $\langle\phi| \mapsto \langle\phi| A$ is linear.

Order and Notation — What is (and isn't) Defined

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$\langle\phi|\, A$ (a bra resulting from the action of A on $\langle\phi|$).

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Relative order matters

$\langle\phi|\, A \neq A\, \langle\phi|$ (the latter is *not* a defined mathematical object).

Operators act **to the right** on kets and **to the left** on bras.

The adjoint operator of a linear operator A

Correspondence between kets and bras

Let $|\psi\rangle$ be arbitrary. The operator A associates with it the ket

$$|\,\psi'\,\rangle \;=\; A\,|\psi\rangle.$$

By the ket–bra correspondence, to $|\,\psi'\,\rangle$ there corresponds a bra $\langle\psi'\,|$.

The adjoint operator of a linear operator A

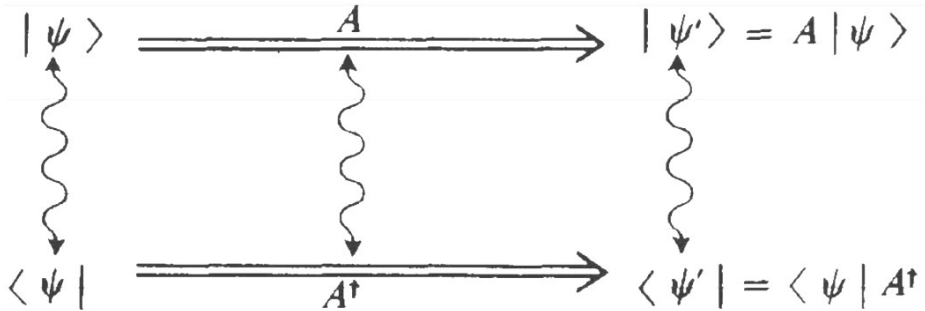
Definition

The adjoint (Hermitian conjugate) $A^\dagger$ is defined by the rule

$$\boxed{|\,\psi'\,\rangle = A|\psi\rangle \quad \Longleftrightarrow \quad \langle\psi'\,| \;=\; \langle\psi\,|\,A^\dagger}$$

for every $|\psi\rangle$.

Definition of the adjoint via the ket–bra correspondence



Adjoint operator: linearity and definition

Linearity of the adjoint (as an operator)

For bras $\langle\psi_1|, \langle\psi_2|$ and scalars λ_1, λ_2,

$$\boxed{\left(\lambda_1\langle\psi_1| + \lambda_2\langle\psi_2|\right) A^\dagger = \lambda_1\langle\psi_1|A^\dagger + \lambda_2\langle\psi_2|A^\dagger}$$

(This shows $A^\dagger$ is a linear operator on the space.)

Adjoint operator: linearity and definition

Property of the scalar product

For arbitrary kets $|\psi'\rangle, |\varphi\rangle$,

$$\boxed{\langle\psi'|\varphi\rangle = \langle\varphi|\psi'\rangle^*}$$

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Substitute the adjoint definition :

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Equate the two sides:

$$\boxed{\langle \psi | A^\dagger | \varphi \rangle = \langle \varphi | A | \psi \rangle^*}$$

(When A leaves the bra, it becomes $A^\dagger$ and goes to the right.)

Important relations and notation

Important relationship satisfied by A

For any arbitrary kets $|\psi\rangle$ and $|\varphi\rangle$,

$$\boxed{\langle\psi|A^\dagger|\varphi\rangle \;=\; \langle\varphi|A|\psi\rangle^*}$$

Important relations and notation

Comment about notation

Scalars and operators inside bras:

$$\langle \lambda\psi| \;=\; \lambda^*\langle\psi|, \qquad \langle A\psi| \;=\; \langle\psi|A^\dagger.$$

Remember:

- When a linear operator A is taken *outside* the bra symbol, it must be replaced by its adjoint $A^\dagger$ and placed to the right: $\langle A\psi| = \langle\psi|A^\dagger$.
- Antilinearity of bras in scalars: $\langle \lambda_1\chi_1 + \lambda_2\chi_2| = \lambda_1^*\langle\chi_1| + \lambda_2^*\langle\chi_2|$.

Properties of the adjoint operator

Using the previously derived relations, it is easy to show that:

$$\begin{aligned} (A^\dagger)^\dagger &= A, \\ (\lambda A)^\dagger &= \lambda^* A^\dagger, \qquad (\lambda \text{ a number}), \\ (A+B)^\dagger &= A^\dagger + B^\dagger. \end{aligned}$$

Properties of the adjoint operator

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Write $|\varphi\rangle$ in the form $|\varphi\rangle = A|\chi\rangle$, setting $|\chi\rangle = B|\psi\rangle$.
Then

$$\langle\varphi|\, AB\, |\psi\rangle^* = \langle\varphi|\, A\, |\chi\rangle^* = \langle\chi|\, A^\dagger\, |\varphi\rangle = \langle\psi|\, B^\dagger A^\dagger\, |\varphi\rangle$$

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$$\boxed{(AB)^\dagger = B^\dagger A^\dagger}$$

Note: the order of the operators changes when taking the adjoint of a product.

Hermitian conjugation: definition and notation

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- Changes the order of the objects.

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Effect of Hermitian conjugation:

- Changes each ket into its corresponding bra, and vice versa.
- Changes the order of the objects.
- Replaces each operator by its Hermitian conjugate.

Hermitian conjugation changes order and type

Let us show that $(|u\rangle \langle v|)^\dagger = |v\rangle \langle u|$

- Applying the adjoint operator relation to the operator $|u\rangle\langle v|$ we find:

$$\boxed{\langle\psi| \left(|u\rangle\langle v|\right)^\dagger |\varphi\rangle \;=\; \left[\, \langle\varphi| \left(|u\rangle\langle v|\right) |\psi\rangle \,\right]^*}$$

Expansion of hermitian conjugation

- Using the property of the complex conjugate of the scalar product:

$$\boxed{\left[\, \langle\varphi| \left(|u\rangle\langle v|\right) |\psi\rangle \,\right]^* = \langle\varphi|u\rangle^* \langle v|\psi\rangle^* = \langle\psi|v\rangle \langle u|\varphi\rangle = \langle\psi| \left(|v\rangle\langle u|\right) |\varphi\rangle}$$

Therefore

$$\boxed{\langle\psi| \left(|u\rangle\langle v|\right)^\dagger |\varphi\rangle = \left[\, \langle\varphi| \left(|u\rangle\langle v|\right) |\psi\rangle \,\right]^*}$$

Rule for Hermitian conjugation (or adjoint)

Result of the operation

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3. Replace the bras by the **kets** associated with them.
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5. Reverse the order of the factors.

Rule for Hermitian conjugation (or adjoint). Examples

If $\lambda \langle u| A |v\rangle |w\rangle \langle \psi|$ is an operator, the adjoint is:

$$(\lambda \langle u| A |v\rangle |w\rangle \langle \psi|)^* = |\psi\rangle \langle w| \langle v| A^\dagger |u\rangle \lambda^* = \lambda^* \langle v| A^\dagger |u\rangle |\psi\rangle \langle w|$$

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Similarly:

$$(\lambda |u\rangle \langle v|w\rangle)^* = \langle w|v\rangle \langle u| \lambda^*$$

Hermitian operators

Rule for Hermitian operators

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Consequences:

- $\langle\psi| A |\varphi\rangle = \langle\varphi| A |\psi\rangle^*$
- $\langle A\varphi|\psi\rangle = \langle\varphi|A\psi\rangle$
- $P_\psi = |\psi\rangle \langle\psi|$ is Hermitian: $P_\psi^\dagger = |\psi\rangle \langle\psi| = P_\psi$
- If A and B are Hermitian, AB is Hermitian only if $[A, B] = 0$